Ansys
2025 R2

Ansys Fluent Simulation Report

Analyst	ShubhamR
Date	4/3/2026 10:33 PM

Table of Contents

System Information

Application	Fluent
Settings	2d, double precision, density-based implicit, SST k-omega
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:44:41 EDT
CPU	AMD Ryzen 7 6800H with Radeon Graphics
OS	Windows

Geometry and Mesh

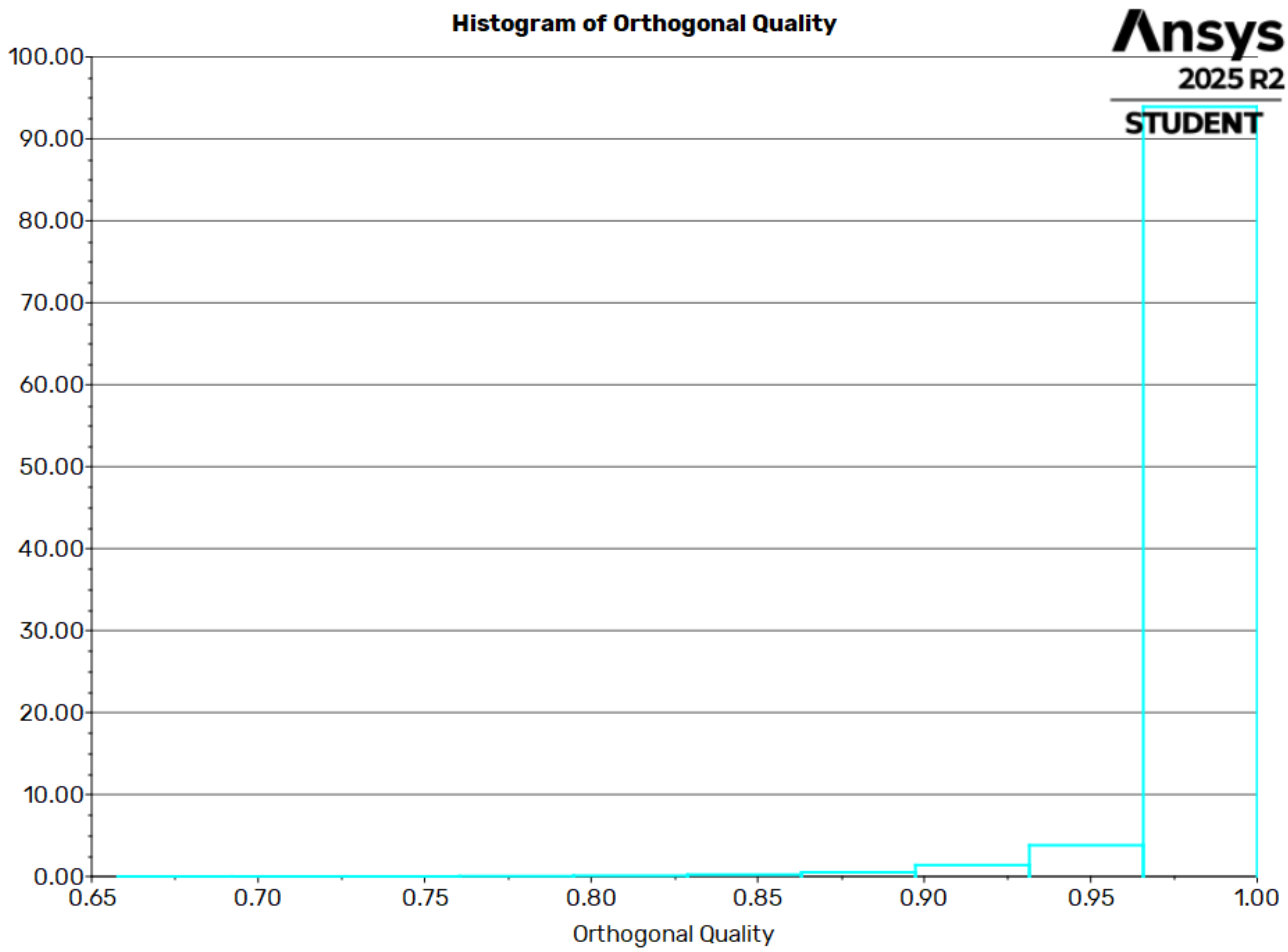
Mesh Size

Cells	Faces	Nodes
38803	58488	19686

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Tri Cell	0.65788031	3.734223

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	SST k-omega turbulence model
Heat Transfer	Enabled

Material Properties

− Fluid	
− air	
Density	ideal gas
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	1.7894e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
− Solid	
− aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

− Fluid	
− fluid	
Material Name	air

Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no

Boundary Conditions

− Inlet	
− inlet	
Reference Frame	Absolute
Gauge Total Pressure [Pa]	792840
Supersonic/Initial Gauge Pressure [Pa]	750000
Total Temperature [K]	300
Direction Specification Method	Normal to Boundary
Build artificial walls to prevent reverse flow?	no
Turbulence Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
− Outlet	
− domain_walls:001	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	101325
Pressure Profile Multiplier	1
Backflow Total Temperature [K]	300
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Acoustic Wave Model	Off
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no

Average Pressure Specification?	no
Specify targeted mass flow rate	no
− domain_walls	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	101325
Pressure Profile Multiplier	1
Backflow Total Temperature [K]	300
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Acoustic Wave Model	Off
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
− outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	101325
Pressure Profile Multiplier	1
Backflow Total Temperature [K]	300
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Acoustic Wave Model	Off
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
− Wall	

– domain_walls:003	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
– domain_walls:002	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
– nozzle_walls	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0

Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1

Reference Values

Area	1 m^2
Density	8.848594 kg/m^3
Depth	1 m
Enthalpy	301929 J/kg
Length	1 m
Pressure	101325 Pa
Temperature	295.2846 K
Velocity	97.424 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	domain_walls:003

Solver Settings

− Equations	
Flow	True
Turbulence	True
− Numerics	
Absolute Velocity Formulation	True
− Under-Relaxation Factors	
Turbulent Kinetic Energy	0.8
Specific Dissipation Rate	0.8
Turbulent Viscosity	1
Solid	1

− Discretization Scheme	
Flow	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
− Time Marching	
Solver	Implicit
Courant Number	3
− Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	6.514 seconds
Iteration	317.28 seconds
AMG	109.081 seconds
Virtual Current Memory	1.41575 GB
Virtual Peak Memory	1.60554 GB
Memory Per M Cell	19.9045

Solution Status

Iterations: 6930

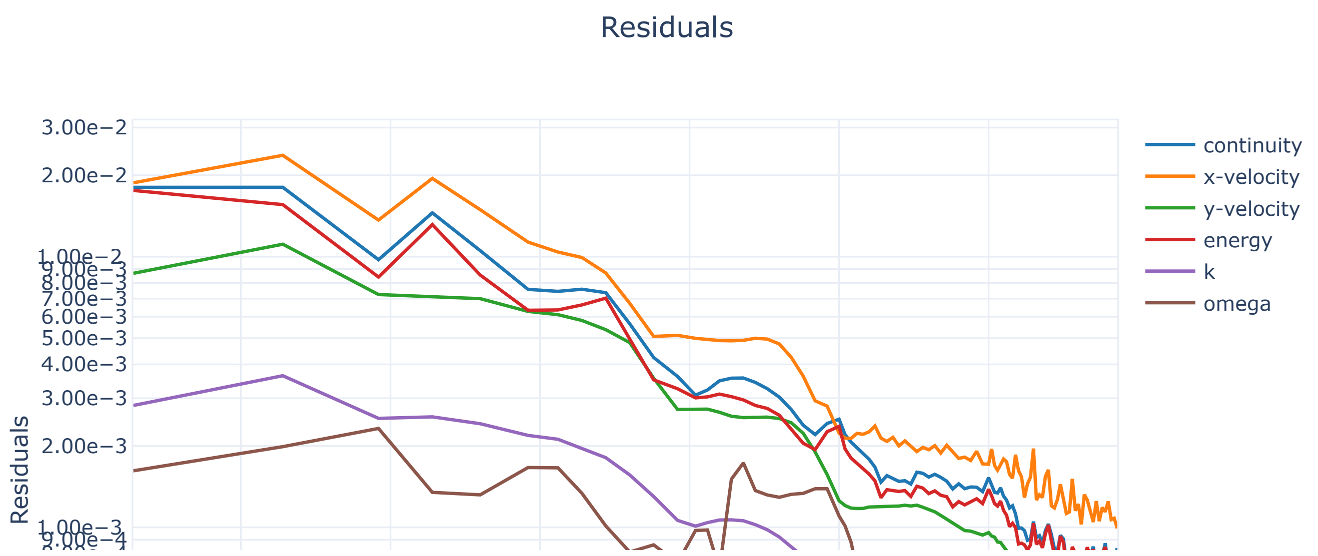
	Value	Absolute Criteria	Convergence Status
continuity	0.0008386507	0.001	Converged
x-velocity	0.0009915328	0.001	Converged
y-velocity	0.0004455444	0.001	Converged
energy	0.0008109147	0.001	Converged
k	0.0001118073	0.001	Converged
omega	0.0001546634	0.001	Converged

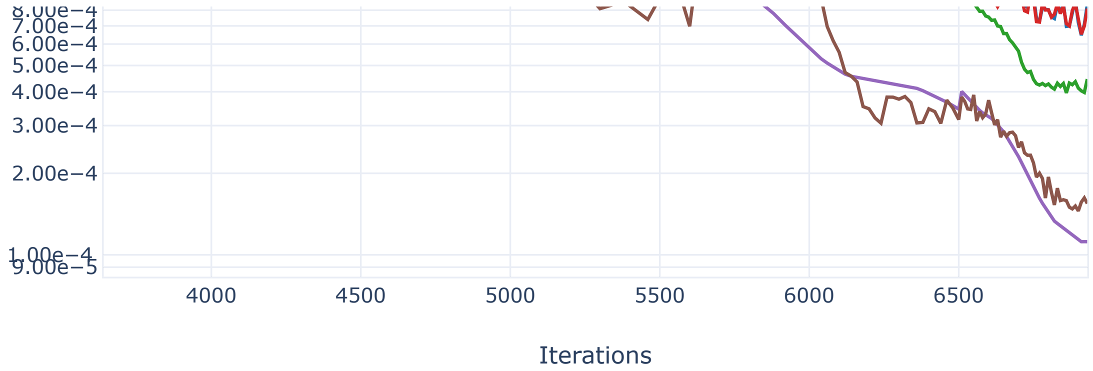
Report Definitions

net-mfr	-39.71697	kg/s
cfl-number	3	

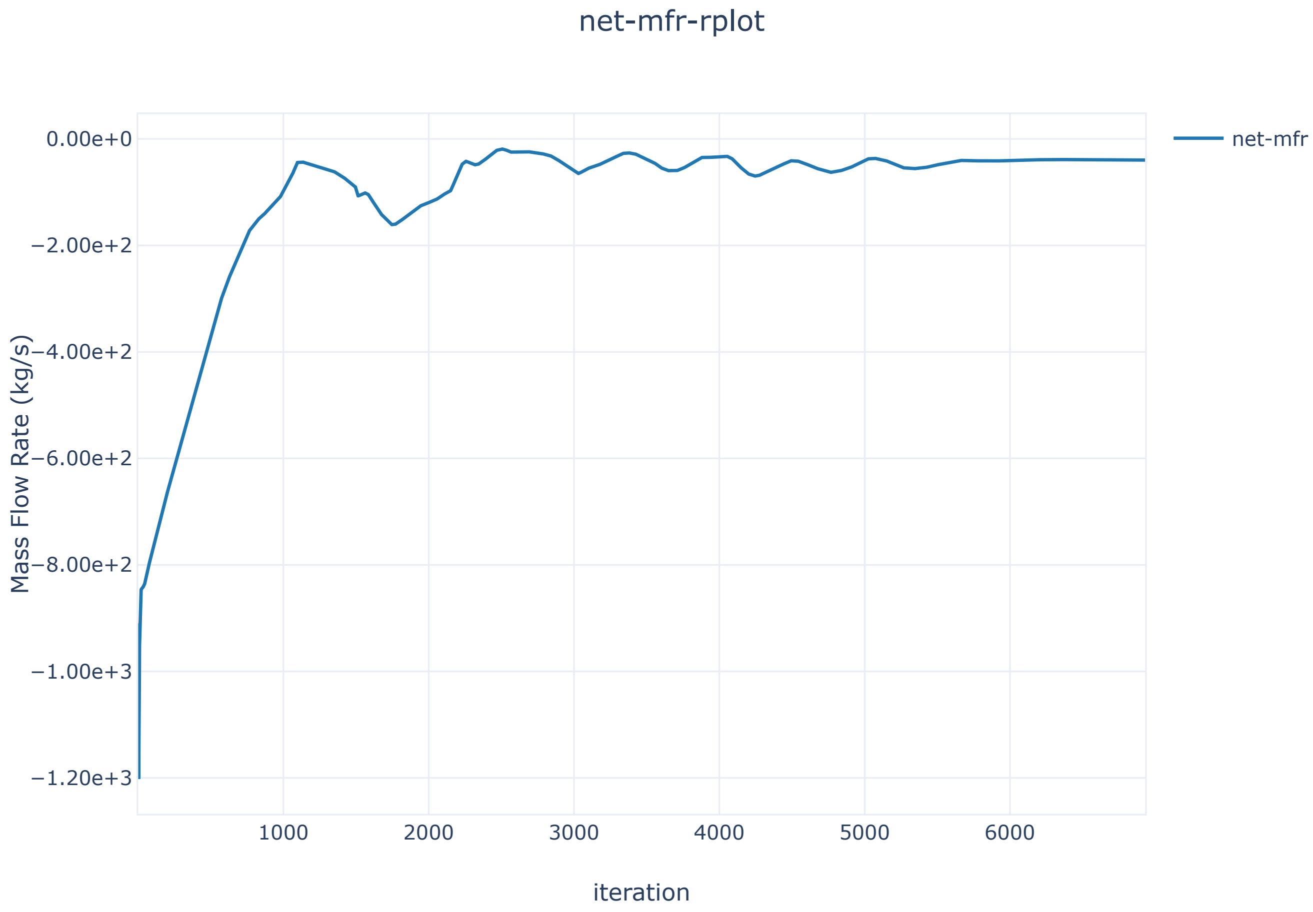
Plots

Residuals





net-mfr-rplot



Contours

contour-3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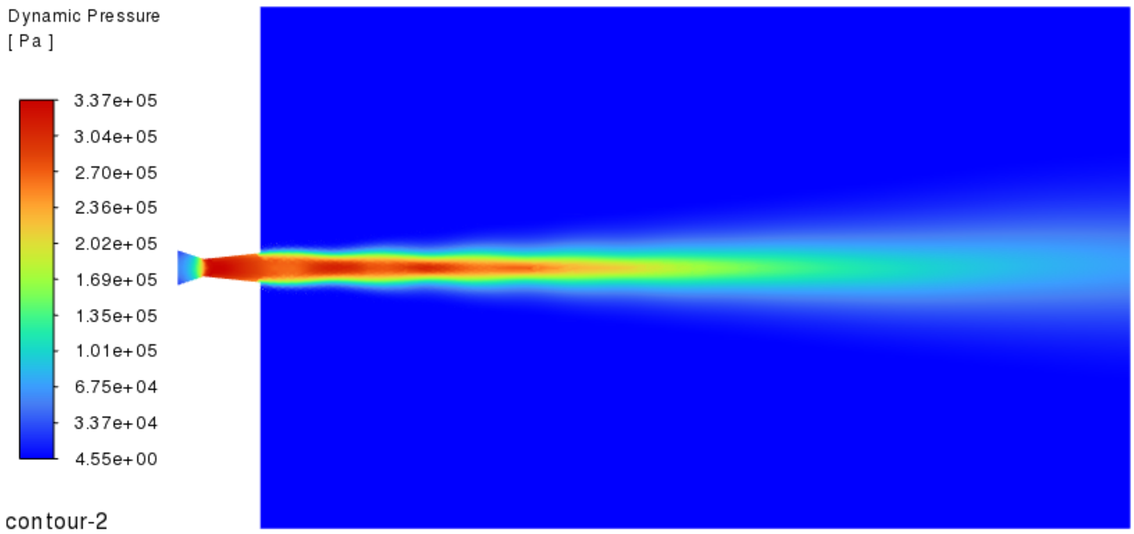
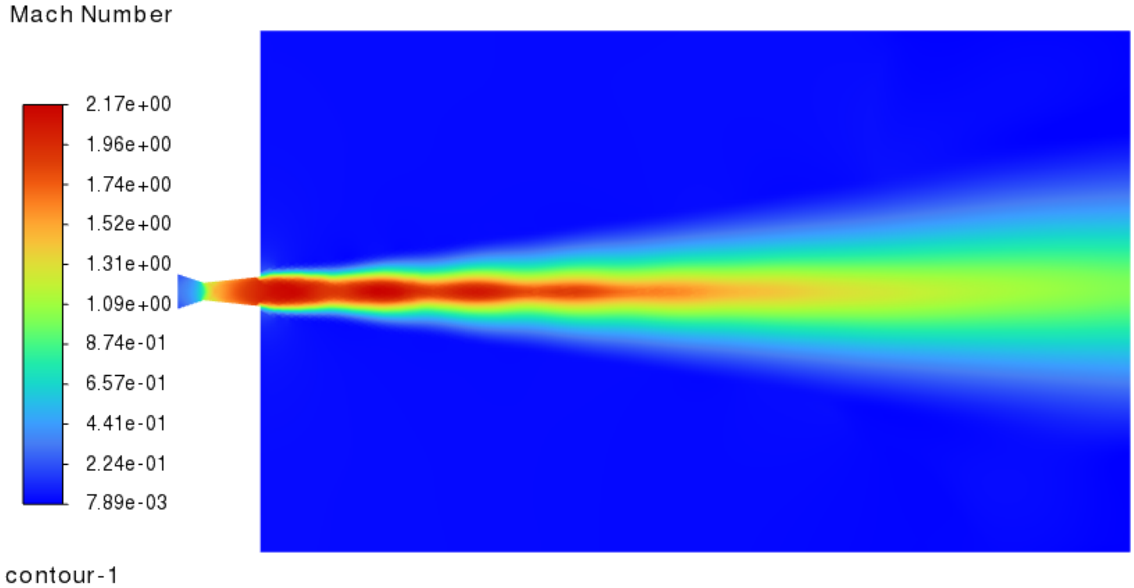
Density
[kg/m^3]
8.92e+00
8.14e+00
7.36e+00
6.59e+00
5.81e+00
5.03e+00
4.25e+00
3.47e+00
2.70e+00
1.92e+00
1.14e+00
contour-3

contour-2

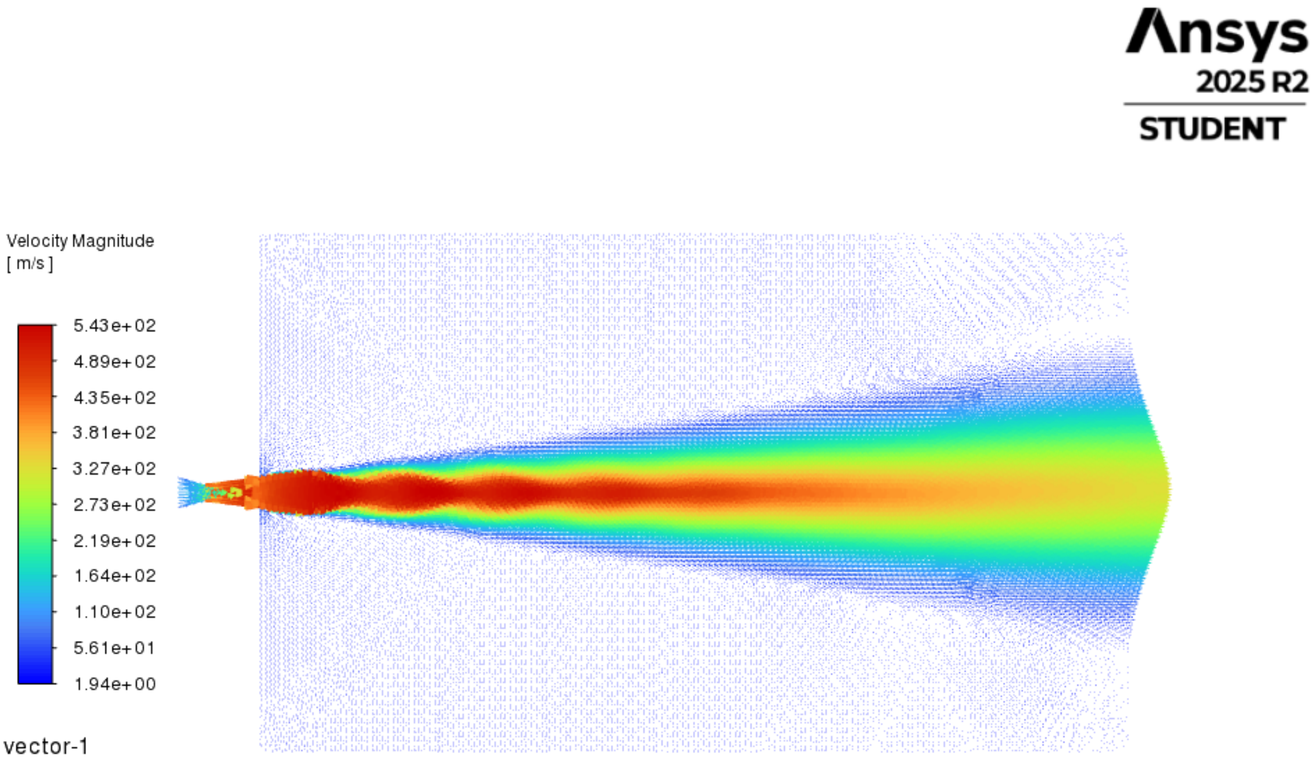

Dynamic Pressure
[Pa]
3.37e+05
3.04e+05
2.70e+05
2.36e+05
2.02e+05
1.69e+05
1.35e+05
1.01e+05
6.75e+04
3.37e+04
4.55e+00
contour-2

contour-1


Mach Number
2.17e+00
1.96e+00
1.74e+00
1.52e+00
1.31e+00
1.09e+00
8.74e-01
6.57e-01
4.41e-01
2.24e-01
7.89e-03
contour-1

Vectors

vector-1



Scenes

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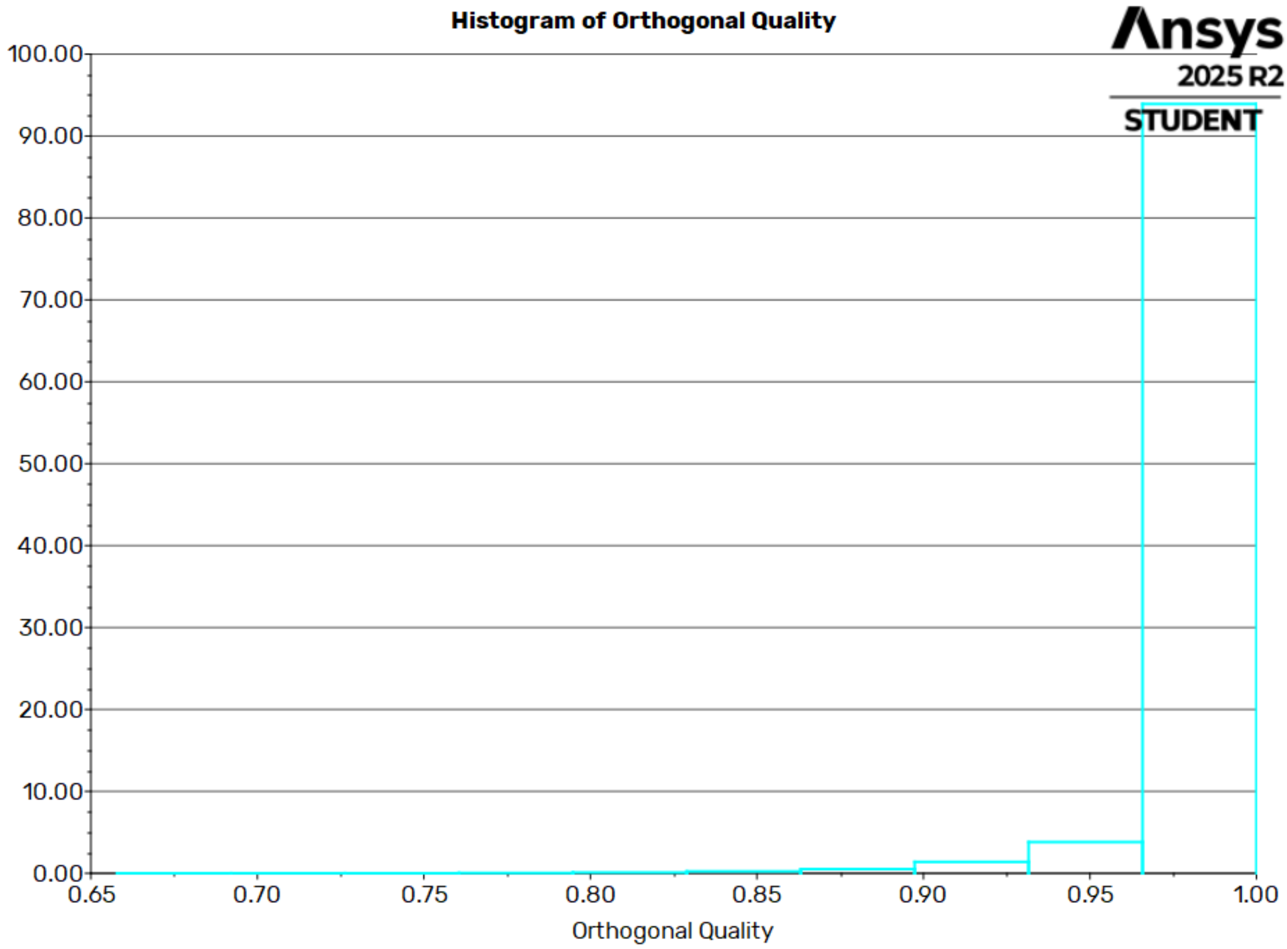
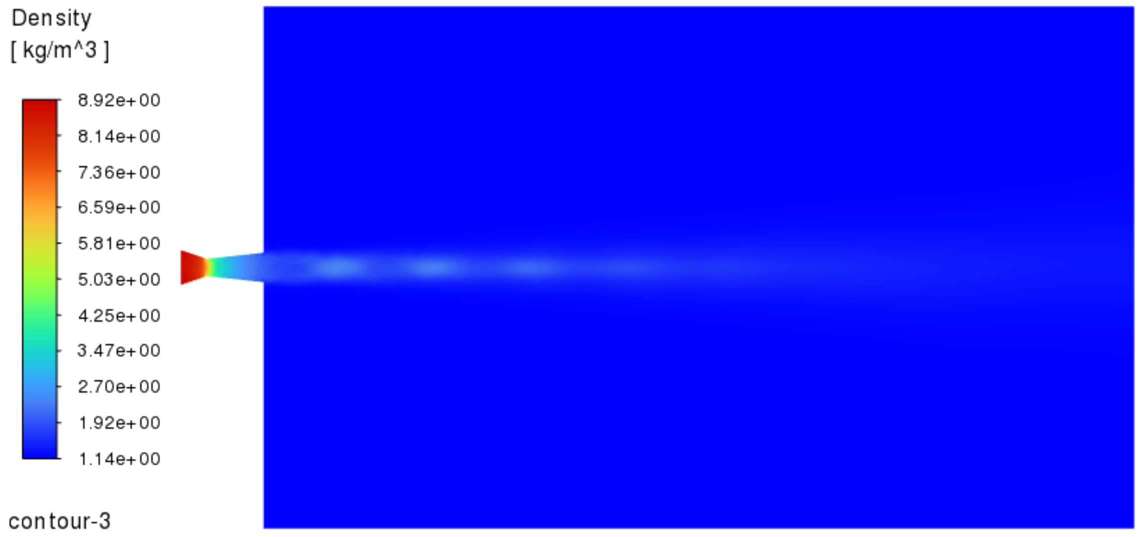
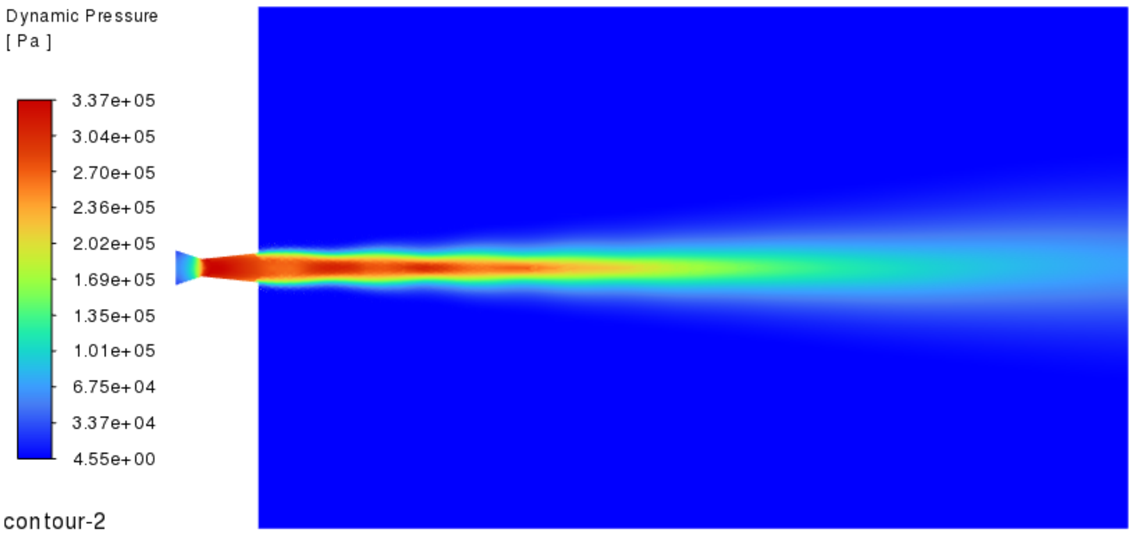

Histogram of Orthogonal Quality
100.00
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Orthogonal Quality


Density
[kg/m^3]
8.92e+00
8.14e+00
7.36e+00
6.59e+00
5.81e+00
5.03e+00
4.25e+00
3.47e+00
2.70e+00
1.92e+00
1.14e+00
contour-3



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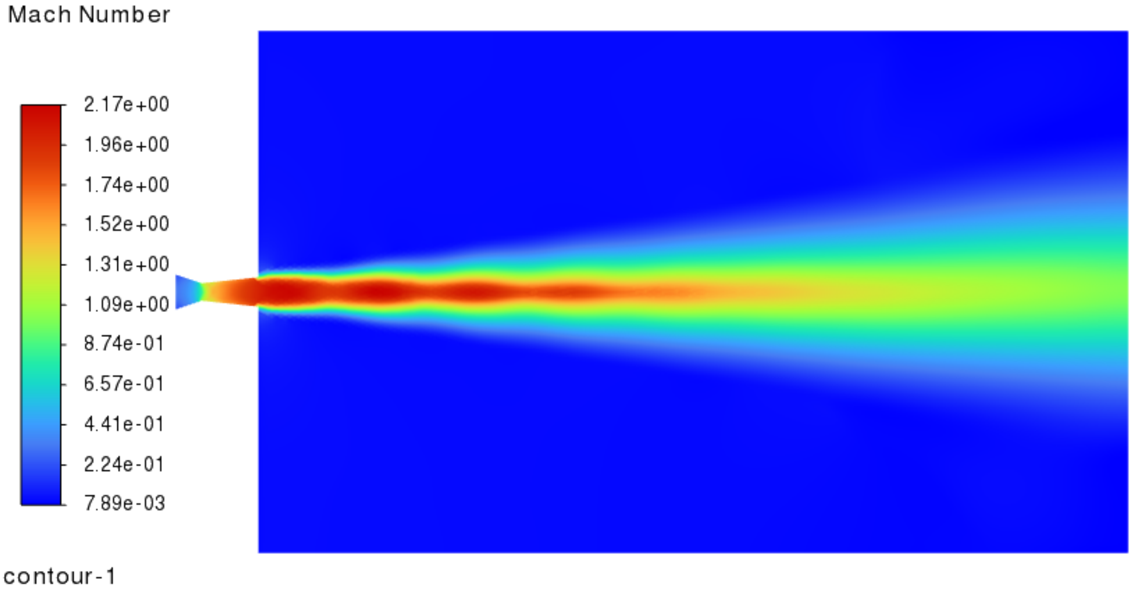

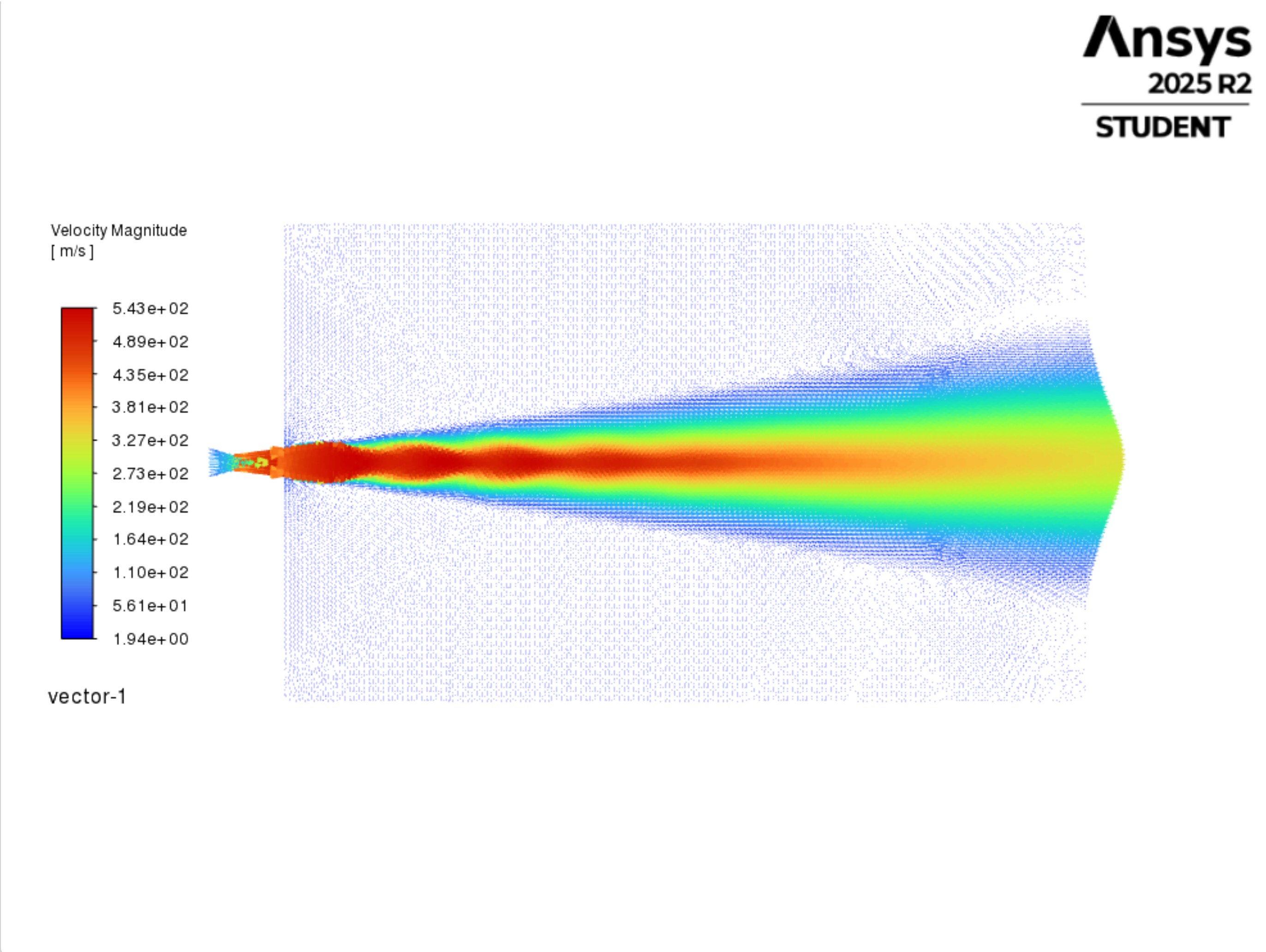
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Velocity Magnitude
[m/s]
5.43e+02
4.89e+02
4.35e+02
3.81e+02
3.27e+02
2.73e+02
2.19e+02
1.64e+02
1.10e+02
5.61e+01
1.94e+00
vector-1